Detecting and Characterizing Back-Button Hijacking on the Web

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Abstract

The Back button lets users return to pages they have already visited. Some websites interfere with it, so that pressing Back does not return the visitor to the page they came from, a practice known as *back-button hijacking*. We present the first systematic large-scale measurement of this behavior, defining it by the outcome of backward navigation rather than by the APIs a page calls. We build an automated detector from a manual examination of 500 websites and evaluate it on 300 held-out websites, where it reaches 75.0% precision and 100% recall on homepages and 85.7% precision and recall on internal pages. Applying it to the CrUX top 10,000 from eight geographic vantage points yields 80,000 measurements, flags 417 of 9,090 testable websites and confirms 364 of them (4.00%) against recorded screenshots, a precision of 87.3%. Almost a third of hijacking measurements occur only on internal pages, and 79.3% of hijacking pages have already planted an extra history entry at load, against 7.9% of benign pages. Overall prevalence is similar at every vantage point, but individual websites are not consistent across them. Our measurements postdate Google's enforcement in June 2026 and establish a post-enforcement baseline.

CCS Concepts

• **Security and privacy → Privacy protections**.

Keywords

Back-Button Hijacking, Browser Navigation, Dark Patterns, Web Measurement, Browser Security

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1 Introduction

The Back button is a fundamental feature of web browsers, allowing users to return to previously visited pages. Web browsers maintain a session history that records navigation between pages, enabling users to navigate backward through their browsing history [29]. Users expect the Back button to return them to the page they previously visited. However, websites can interfere with this behavior, preventing users from navigating backward as expected.

This practice, known as *back-button hijacking*, occurs when websites manipulate browser navigation to prevent users from returning to the previous page. Instead, users may remain on the same page, be redirected to unexpected destinations, or encounter unsolicited advertisements [19]. Each of those outcomes has been studied on its own: malicious and injected advertising [28, 30, 31] and malicious URL redirection [12, 13, 27] both have measurement literatures, as does the session history as an attack surface [21, 25]. What distinguishes back-button hijacking is that the navigation the user asked for does not happen at all. Such behavior interferes with users' control over browser navigation and can be considered a form of deceptive web design, commonly referred to as a *dark pattern* [7, 8, 16, 17], a class of design that has been measured across shopping sites, consent dialogues and mobile applications [10, 16, 20], and that affects automated agents as well as people [5]. Back-button hijacking has also attracted attention from browser vendors, search engines and the trade press [9]. Chromium introduced a mitigation that skips certain history entries created without user interaction [2]. More recently, Google announced in April 2026 that back-button hijacking violates its spam policies, with enforcement beginning on June 15, 2026 [19]. Reporting at the time put the practice among publishers and among the content-recommendation platforms that offer it to them, while noting that how many publishers deploy it is unknown [9]. Despite these efforts and prior research on dark patterns, to the best of our knowledge, no academic study has systematically measured back-button hijacking at scale, leaving its prevalence and characteristics poorly understood.

Identifying back-button hijacking presents several challenges. First, websites commonly use the History API to manage legitimate navigation [29], making it difficult to distinguish back-button hijacking from normal browser history manipulation based on API calls alone. Second, back-button hijacking is triggered by backward navigation, requiring measurements to establish a previous history entry and simulate user interactions to observe the behavior. Third, the behavior must be observed through an automated crawler, and crawlers are themselves an imperfect instrument: measurement tools are detectable and are served different content than ordinary visitors [1, 14], results vary with the browser and automation stack [24, 26], and web measurements are difficult to reproduce [3]. Because our signal is a navigation that fails to occur, any of these conditions produces a record that resembles hijacking.

In this paper, we develop an automated approach to identify back-button hijacking at scale by analyzing browser navigation behavior and history manipulation, in the tradition of large-scale automated web measurement [4]. We sample from the Chrome User Experience Report (CrUX) [6], which is built from the traffic of opted-in Chrome users and has been characterized across devices and countries [23]; the choice of popularity list is itself known to affect what a measurement finds [15]. Starting with 500 randomly selected websites from CrUX, we use 200 websites to develop our detection approach and the remaining 300 to evaluate its performance,

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achieving 75.0% precision and 100% recall on homepages, and 85.7% precision and recall on internal pages. We then apply our approach to the CrUX top 10,000 websites from eight geographic vantage points, identifying 417 websites exhibiting back-button hijacking among 9,090 testable websites (Section 3). Our main findings are summarized as follows:

- Back-button hijacking is confirmed on 364 of the 9,090 testable websites (4.00%). Of the 1,738 hijacking measurements, 60.2% are homepage-only, 29.1% internal-page-only and 10.7% both (§ 4.1).
- CrUX ranks among hijacking websites follow the distribution of all testable websites, so the behavior is not concentrated at any popularity rank (§ 4.2).
- Adult (18.7%), news (13.2%), gambling and gaming (9.6%) and shopping (9.6%) hold about 51% of the confirmed hijackers. By rate, adult is highest: 5.72% against 4.16% overall, and the only category that stays above the overall rate when domains are collapsed to operators (§ 4.3).
- Same-page trapping is the most common technique, and the dominant mechanism precedes the first press: 79.3% of hijacking pages plant an extra history entry at load, against 7.9% of benign pages. We also observed multi-step backward navigation, unexpected navigation destinations and advertisement vignettes, the last in 1 of the 1,738 measurements, so a criterion keyed on advertising would have detected almost nothing (§ 4.4).
- Prevalence is similar at every vantage point, which conceals two populations: of the 247 confirmed websites hijacking at one or more of the eight, 37.7% hijack at all of them and 23.5% at exactly one. Per-vantage rates alone would have reported a null result (§ 4.5).

Finally, we discuss the implications of our findings for browser security, website operators, and policy enforcement, as well as the limitations of our measurement approach (Section 5).

2 Background and Related Work

2.1 Browser History and Navigation

Web browsers maintain a session history for each tab, recording the sequence of visited pages [29]. The Back button allows users to navigate to previous entries in this history.

The History API provides two methods for manipulating session history without loading a new document: `history.pushState()`, which adds a new history entry, and `history.replaceState()`, which modifies the current entry. These methods are commonly used by single-page applications to manage navigation without reloading the page. In addition, the `popstate` event allows websites to respond when users navigate between history entries within the same document.

While these mechanisms support legitimate navigation, they can also be misused to interfere with the Back button. For example, a website can insert additional history entries or respond to a Back press by pushing another entry, preventing users from returning directly to the previous page. Chromium introduced a mitigation against back-button hijacking by skipping certain history entries created without user interaction [2]. However, this mitigation does not cover all forms of navigation manipulation, including certain history changes following user interaction.

2.2 Back-Button Hijacking and Related Work

Back-button hijacking occurs when a website interferes with browser navigation, preventing users from returning to the previously visited page when pressing the Back button. This behavior may involve redirecting users to unexpected pages, displaying unsolicited advertisements, or preventing backward navigation altogether [19]. Based on this description, we define back-button hijacking as follows:

> *A website exhibits back-button hijacking if pressing the browser's Back button does not immediately return the user to the previously visited page due to website-induced navigation manipulation.*

Our definition focuses on the outcome of backward navigation rather than specific JavaScript APIs or the presence of advertisements. We consider cases requiring multiple Back presses to return to the previous page as back-button hijacking. The detailed classification criteria are described in Section 3.2.

Back-button hijacking has been discussed in industry reports [9] and is closely related to deceptive web practices studied in prior research. Existing studies have investigated dark patterns that manipulate user decisions [7, 16], as well as malicious advertising and unwanted web navigation [31]. However, these studies do not specifically investigate the prevalence and characteristics of back-button hijacking.

In April 2026, Google announced that back-button hijacking would become an explicit violation of its spam policies, with enforcement beginning on June 15, 2026 [19]. Websites engaging in this behavior may face manual spam actions or automated ranking demotions. Google also noted that back-button hijacking may originate from third-party libraries or advertising platforms. Our measurements were taken after that date, so they establish a post-enforcement baseline; § 5.4 explains why a before-and-after comparison is not available from our data.

Navigation abuse has been measured from several angles. Khodayari et al. studied client-side open redirects across the Tranco top 10K, finding 20.8K vulnerabilities on 623 sites and showing that more than 11.5% could be escalated to XSS, CSRF, or information leakage [12]. Koop et al. crawled the Alexa top 50K and analysed 108,435 redirect chains, reporting that 11.6% of sites use one of the top 100 redirectors to store first-party cookies that ordinary blockers miss [13]. Iqbal et al. built a classifier for advertising and tracking request chains [11].

Attacks on the session history itself are a separate line. Smith et al. describe four history sniffing attacks that read which URLs a user has visited, one of which exfiltrates at roughly 3,000 URLs per second [25]. Reverse tabnabbing lets a page opened with `target="_blank"` rewrite the page that opened it, which browsers now mitigate by treating such links as implicitly `noopener` [21]. Both read or write across a navigation boundary; back-button hijacking leaves the boundary intact and removes the control that crosses it.

Work on consent interfaces measures a related gap. Nouwens et al. found that 11.8% of consent platforms on UK sites met a minimal reading of European law [20], and Matte et al. found consent

registered before any choice was made on a quarter of the sites they audited [18]. Both describe an interface reporting an action the user did not take; we describe a control that does not perform the action the user did take.

In contrast to prior studies on malicious redirects [12], browser history attacks [25], and deceptive web interfaces [16], our work focuses on back-button hijacking, where websites interfere with users' ability to navigate backward. To the best of our knowledge, we present the first systematic large-scale measurement of this behavior. We develop an automated detection approach and characterize its prevalence, implementation techniques, and variations across websites and geographic locations.

3 Back-Button Hijacking Identification

3.1 Ground-Truth Collection

3.1.1 Website selection. We randomly selected 500 websites from the Chrome User Experience Report (CrUX) [6], a monthly updated public dataset maintained by Google that aggregates anonymized real-world browsing data from opted-in Chrome users, making it suitable for studying back-button hijacking in the wild [1]. Before sampling, we extracted and normalized domain names from the CrUX origins by removing the scheme (e.g., `https://`), the `www.` prefix, port, and path, and converting them to lowercase. We discarded invalid entries and removed duplicate domains to avoid overrepresentation.

We randomly divided the 500 websites into a development set of 200 and a held-out evaluation set of 300. We used the development set to identify detection errors and refine our methodology, and the evaluation set to independently evaluate the performance of our detection approach (see § 3.3).

3.1.2 Manual examination. We manually examined both the homepage and an internal page of each website. For each website, we opened a fresh incognito browser session, visited `google.com`, and then navigated to the target homepage to establish a previous entry in the browser history. We then waited for the page to finish loading, scrolled to the bottom, clicked on an empty area without a link[2], and pressed the browser's Back button to record the navigation outcome. We also examined an internal page by selecting a link from the homepage that led to the same website, excluding links to login account, and other non-content pages. We prioritized links likely to lead to articles or other content based on their URL structure and link text. We then clicked the selected link and followed the same examination procedure used for the homepage.

3.1.3 Ground-Truth labeling. Following the definition in § 2.2, we classify back-button hijacking based on the outcome of backward navigation. We define the *referrer* as the page from which the user navigated to the target page. For homepage measurements, the referrer is `google.com`, while for internal-page measurements, it is the website's homepage. Under normal navigation, pressing Back should return the user to the corresponding referrer. We label a page as hijacking if pressing Back fails to return to the referrer, including cases where multiple Back presses are required or the browser navigates to an unexpected page.

We classify a website as hijacking if either its homepage or internal page exhibits this behavior. Our classification is based on navigation outcomes, regardless of whether advertisements are displayed during backward navigation.

[1]We used the CrUX top-1,000,000 origins snapshot dated April 14, 2026.

[2]Clicking an empty area supplies a user gesture, which changes how the browser treats any navigation the page subsequently performs. We observe that without a gesture, a scripted navigation might not trigger.

Table 1: Hijacking confirmed by manual review of 500 CrUX websites, by mechanism and page. The Home, Internal and Both columns partition Sites, and the mechanism rows partition the total.

Mechanism	Sites	HomePage	Internal page	Both
Same-page trapping	22	17	2	3
Multi-step backward navigation	2	1	1	0
Advertisement vignette	1	0	1	0
Unexpected navigation destinations	1	1	0	0
No mechanism identified	7	4	2	1
Total confirmed	**33**	**23**	**6**	**4**

3.1.4 Behaviors observed. We summarize our manual examination in Table 1.The *no mechanism identified* row counts websites where backward navigation plainly failed to return us to the referrer, and we could see that it failed, but we could not attribute the failure to a specific technical mechanism with confidence. These are confirmed hijacking websites: the uncertainty concerns the implementation, not whether the behavior occurred. We record them as a category of their own rather than assigning a mechanism we could not establish. We confirmed back-button hijacking on 33 of the 500 websites (6.6%). Of these, 23 exhibited hijacking only on the homepage. Six of the 33 confirmed websites (18.2% of websites) exhibited hijacking only on an internal page, while backward navigation worked normally on their homepages. We therefore measure an internal page as well as the homepage.

We identified several forms of back-button hijacking. The most common was *same-page trapping*, found on 22 websites, where pressing Back left users on the same page instead of returning to the previous page. Other mechanisms included *multi-step backward navigation*, requiring multiple Back presses to return; *advertisement vignettes*, displaying full-screen advertisements during backward navigation; and *unexpected navigation destinations*, redirecting users to a different page.

3.2 Back-Button Hijacking Identification

3.2.1 Navigation procedure. We developed an automated browser-based approach to identify back-button hijacking by reproducing the navigation procedure described in Section 2.2. For each website, we examined both its homepage and an internal page using a new, clean browser profile with no cookies, cache, or browsing history. We first visited `google.com` before navigating to the target homepage to establish a previous entry in the browser's session history. We wait for `document.readyState` to reach `complete`, indicating that the page has finished loading. We then monitor the URL until it remains unchanged for two consecutive seconds to account for asynchronous navigation and redirects after page loading. To prevent cookie consent banners from obstructing webpage interactions, we integrated *BannerClick* [22] to automatically detect and

accept these banners[3]. If accepting consent triggers an unexpected navigation, we restore the original page and mark the measurement as contaminated. We then scroll through the page to account for scripts that may activate after user interaction. Finally, we press the Back button up to three times, recording the navigation outcome after each press. Based on our manual examination, we chose this limit to capture cases where websites insert one or two additional history entries, requiring multiple Back presses to return to the referrer. Figure 1 illustrates an example where two additional history entries require three Back presses to reach the referrer.

3.2.2 Internal page selection. For each website, we select one internal page from the links available on its homepage. We examine the first 1,500 links and consider only HTTP(S) URLs within the same registrable domain. We exclude fragment-only links, homepage-equivalent URLs, and non-content pages such as login, registration, account, checkout, privacy, and contact pages.

We rank the remaining links based on their URL structure and link text, prioritizing content-related paths (e.g., `/article/`, `/news/`, `/blog/`, `/post/`, and `/product/`), numeric URL components, deeper paths, and descriptive link text. We assign lower scores to generic links such as "more" or "view all". For links without visible text, we use the `aria-label`, `title`, or image `alt` attributes to obtain their descriptions. We then select the highest-ranked eligible link, excluding social media links and non-document resources such as PDFs and images.

We access the selected internal page by clicking its anchor element rather than navigating directly to its URL. This ensures that website-specific click handlers, which may manipulate browser history, are executed. If fewer than five links are available after a second attempt, we classify the page as a rendering failure and exclude it from the internal-page measurement.

3.2.3 Browser Instrumentation. We inject an observer before website scripts execute to monitor calls to `history.pushState` and `history.replaceState`, `popstate` events, window openings, and navigation timing.

Cross-document navigation. The observer is initialized separately for each document, and its counters are reset when a new document is loaded. The observer is initialized separately for each document, and its counters are reset when a new document is loaded. Comparing counters before and after a cross-document navigation would therefore produce incorrect results. To address this, we assign a unique identifier to each observer and compute changes in event counts only when the document identifier remains unchanged. Otherwise, we mark the comparison as unavailable.

History-entry replacement. In addition, certain navigation methods, such as `location.replace`, cannot be intercepted by overriding the `Location` prototype because their properties are defined directly on the instance. We therefore use navigation timing and browser history length to infer possible history-entry replacement, rather than relying solely on JavaScript API interception.

3.2.4 Classification criteria. We classify each page based on its recorded backward navigation sequence, following the definition in Section 2.2. Before assigning a verdict, we check for measurement artifacts that may interfere with backward navigation, including connection failures, bot-verification pages, authentication redirects, and incomplete client-side rendering. We also exclude internal-page measurements when the selected link opens a new window or tab while backward navigation works normally in the original tab. Measurements affected by these conditions are excluded from classification, while those without a recorded Back-navigation sequence are labeled as *no measurement*. For valid measurements, we determine the first Back press at which the browser returns to the referrer. A page is classified as *benign* if the referrer is reached on the first press, and as *hijacking* if additional presses are required or the referrer is not reached within three presses (see Figure 2). We also record the observed hijacking mechanism, when identifiable, for subsequent analysis.

[3] *BannerClick* achieves high accuracy (around 99%) in automating banner detection and interaction and supports a wide range of banner implementations.

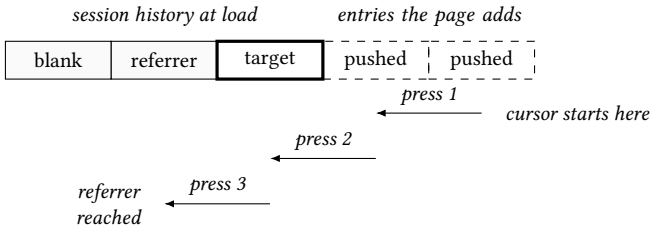


Figure 1: History entries inserted by the target site. The first two Back presses traverse them; the third reaches the referrer.

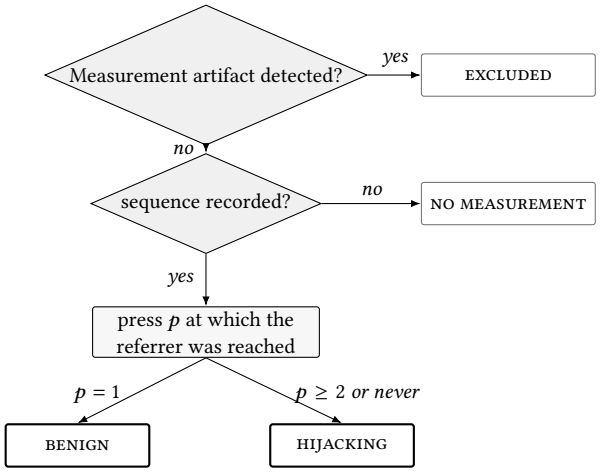


Figure 2: A back sequence classification. p is the first press that reaches the referrer: benign if $p = 1$, hijacking if $p \geq 2$ or never.

URL Comparison. We normalize URLs before comparing navigation destinations by converting hostnames to lowercase, removing tracking parameters, and sorting the remaining query parameters. We also consider URL fragments, since fragment changes may indicate history manipulation but can also result from legitimate single-page application navigation. We therefore make fragment sensitivity configurable and record both comparison outcomes.

Table 2: Detector performance on 300 held-out websites, with Wilson 95% intervals. Unmeasurable domains are excluded.

Page	TP	FP	FN	TN	Precision	Recall
Homepage	18	6	0	230	0.750 [0.55, 0.88]	1.000 [0.82, 1.00]
Internal page	6	1	1	132	0.857 [0.49, 0.97]	0.857 [0.49, 0.97]

3.3 Evaluation

We evaluated our detection approach using the 300 held-out websites that were not used during detector development (§ 3.1). We compared the automated classification results with the manually assigned labels for both homepages and internal pages. A page was considered positive if it was classified as exhibiting back-button hijacking. Measurements that could not be completed were excluded from the evaluation, leaving 254 homepages and 140 internal pages.

Table 2 presents the detection performance. For homepages, our approach achieved a precision of 0.750 and a recall of 1.000. For internal pages, both precision and recall were 0.857. However, these results are based on only 18 positive homepages and 7 positive internal pages, resulting in relatively wide confidence intervals.

We note that our evaluation is limited to the navigation scenarios covered by the manual examination. Since both manual and automated measurements follow the same procedure, they may miss back-button hijacking triggered under different conditions, such as returning visits, mobile browsing, or longer user interactions. Consequently, our measurements may underestimate the prevalence of back-button hijacking across broader browsing scenarios.

Main sources of error. Our evaluation identified seven false positives and one false negative across homepages and internal pages. False positives may arise from navigation failures unrelated to back-button hijacking, such as network errors, bot-verification pages, authentication redirects, or incomplete page rendering. Although our approach includes checks for these measurement artifacts (§ 3.2), some cases remain difficult to distinguish from actual hijacking.

3.4 Large-Scale Detection and Validation

We applied our detection approach to the top 10,000 domains from CrUX, randomly shuffled with a fixed seed while retaining their original rankings. Each vantage point used the same domain list to ensure consistent measurements across locations. To investigate geographic differences, we conducted measurements from eight vantage points using Proton VPN[4]: Germany, Sweden, New York and California (United States), India, Japan, Vietnam, and South Korea. Each vantage point processed the domains in batches of 500, using four concurrent browser instances, with each measurement taking approximately 27 hours per vantage point.

We verified the VPN exit location and network organization before each measurement and after every batch to ensure that traffic originated from the intended vantage point. We excluded measurements with unverifiable network exits. Further details on vantage-point validation are provided in Appendix A.

Measurement outcomes. Table 3 summarizes the results of our large-scale crawl. Across eight vantage points, we conducted 80,000 website measurements, of which 66,283 (82.9%) were successfully classified as either hijacking or benign. The remaining 13,717 measurements consisted of 13,573 cases that could not be evaluated due to access restrictions or rendering failures, and 144 errors in our measurement infrastructure.

[4]https://protonvpn.com/

Table 3: Crawl outcomes per vantage point.

Vantage	Hijacking	Benign	Testable	Not measured	Error	Internal-page coverage
Germany	226	8317	8543	1448	9	59.9%
India	208	7518	7726	2260	14	56.8%
Japan	225	8331	8556	1414	30	61.2%
South Korea	202	8025	8227	1765	8	60.8%
Sweden	234	8340	8574	1417	9	60.1%
USA (CA)	235	8191	8426	1525	49	59.9%
USA (NY)	201	7912	8113	1873	14	59.8%
Vietnam	207	7911	8118	1871	11	61.0%
Total	**1738**	**64545**	**66283**	**13573**	**144**	**59.9%**

The proportion of websites that could not be measured ranged from 14.1% to 22.6% across vantage points. We also obtained valid internal-page measurements for 56.8% to 61.2% of websites, depending on the vantage point. The remaining websites either lacked an eligible internal link or encountered rendering failures.

Detected Hijacking Websites. Our approach identified back-button hijacking in 1,738 of the 66,283 testable measurements (2.62%). Across all vantage points, we identified 417 distinct websites exhibiting back-button hijacking among the 9,090 websites successfully measured from at least one vantage point (4.59%). We validated these results in two stages. First, we re-derived the classification of all 417 websites from their recorded Back-navigation sequences and found no inconsistency between the recorded outcomes and the assigned labels; this checks the classifier against its own input and not against the web. Second, one author reviewed all 417 websites against the before-and-after screenshots captured at each Back press, confirming 364 and overturning 53. For 32 of the 364 no screenshot was recorded, so they were never inspected and we count them with the confirmed set, giving a precision of 87.3% (95% CI 83.7–90.2); counting them against the detector instead gives 79.6%. Each website was reviewed at the vantage point whose record carried the strongest mechanism evidence, so this figure is an upper bound on precision across all vantage points. The analyses in § 4.3 use the 364 confirmed websites rather than the full detector output.

4 Characterization of Back-Button Hijacking on the Web

In this section, we characterize back-button hijacking across the CrUX top 10,000 websites, focusing on its prevalence, website popularity, categories, hijacking mechanisms, and geographic differences. Unless stated otherwise, we calculate prevalence based on testable measurements, excluding cases where no measurement was possible (Table 3). For website-level analyses, each website is counted only once across all vantage points. Our detector flagged 417 websites, and reviewing all of them against the recorded screenshots confirmed 364 (§ 3.4). Every website-level result in this section is computed over those 364. The review ran once per website, so it cannot be applied to individual measurements; the homepage and internal-page split, the press-depth distribution and the mechanism

Table 4: Distribution of back-button hijacking across homepages and internal pages by vantage point.

Vantage	Homepage	Internal page	Both	Total
Germany	136	63	27	226
India	133	54	21	208
Japan	141	63	21	225
South Korea	108	71	23	202
Sweden	147	63	24	234
USA (CA)	138	68	29	235
USA (NY)	121	58	22	201
Vietnam	123	65	19	207
Total	**1047**	**505**	**186**	**1738**

table therefore remain the detector's output, and we say so where they appear.

4.1 Prevalence

Among the 9,090 websites successfully measured from at least one vantage point, 417 were flagged and 364 (4.00%) confirmed as exhibiting back-button hijacking. Across all 66,283 testable website–vantage-point measurements the detector classified 1,738 (2.62%) as hijacking, a figure that is its output and is not screenshot-verified.

In addition, among the 1,738 measurements classified as hijacking, 1,047 (60.2%) exhibited hijacking only on the homepage, 505 (29.1%) only on an internal page, and 186 (10.7%) on both. Table 4 presents the distribution across vantage points. The 29.1% of measurements seen only on internal pages exceeds the 18.2% of websites in our manually examined dataset, so a homepage-only crawl would miss close to a third of these cases. We obtained valid internal-page measurements for only 56.8–61.2% of domains across vantage points, which may affect the observed distribution.

To examine the impact of our classification criteria, we analyze the number of Back presses required to return to the referrer. As shown in Figure 3a, among the 1,924 page-level hijacking cases, the referrer was not reached within three presses in 872 cases (45.3%), while 901 (46.8%) required two presses and 151 (7.8%) required three presses. Excluding cases where the referrer was reached after two or three presses would reduce the number of detected page-level hijacking cases by 54.7%.

4.2 Website Ranking

We examine the popularity of websites exhibiting back-button hijacking based on their CrUX rankings. Figure 3b shows the empirical cumulative distribution function (ECDF) of the ranks of the 364 confirmed hijacking websites compared with all 9,090 testable websites. We observe that the two distributions closely follow each other across the top 10,000 ranks, suggesting that back-button hijacking is not concentrated among websites of a particular popularity rank.

4.3 Website Categories

Classification. Hand-coding does not scale to 9,090 domains, so we classified the corpus with a large language model given each site's hostname and, where the site could be fetched, its page title and metadata. The model assigned one of seventeen categories to 7,928 of the 9,090 testable domains (87.2%) and declined on the rest. To measure how far it stands in for a human coder we compared it against the 417 domains we had already coded by hand: the two agree on 84.7% of the 378 the model answered. Agreement is 88.7% on the domains it marked high confidence and 55.6% on those it marked medium. Domains it declined are excluded from both the numerator and the denominator below, which would otherwise be diluted by domains nobody classified.

Verified positives. The rates here use the 364 positives confirmed by screenshot review (§ 3.4), of which 330 carry a model category; the remaining 34 are excluded from this subsection alone, since a domain with no category cannot enter a per-category rate. The denominator is every testable domain and is not verified, since a domain the detector called benign was never examined. These are therefore confirmed hijackers per testable domain, and recall remains unmeasured.

Composition. Figure 3c gives the category breakdown of the 364 confirmed hijackers. Adult sites account for 18.7%, news for 13.2%, gambling and gaming for 9.6%, and shopping for 9.6%. Those four hold about 51% of the confirmed cases, and the classifier declined on 34 (9.3%).

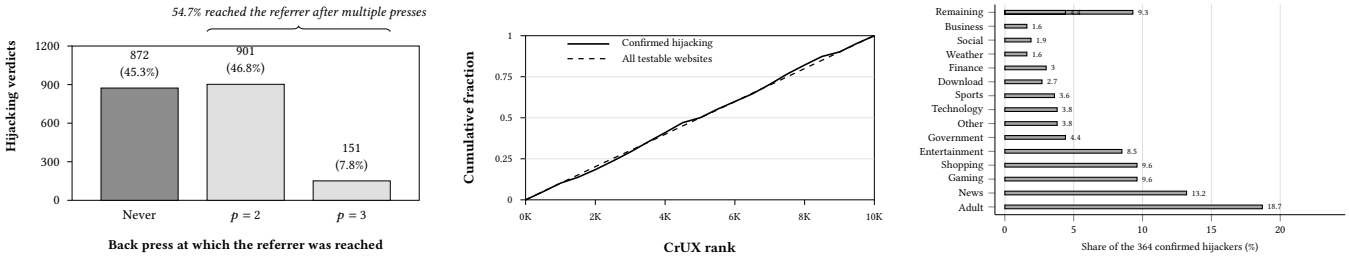
Rate by category. A composition says which categories hijackers fall into, not which categories hijack; the two differ by how common each category is. Against an overall rate of 4.16% within the 7,928 classified domains, adult sites are the highest of the categories with enough domains to compare, at 5.72%, and the only one that stays above the overall rate when domains are collapsed to operators by registrable label, where 6,379 units and 270 confirmed hijackers leave adult at 5.76% against 4.23%. News follows at 5.41% and gambling and gaming at 4.53%. Weather is nominally highest at 8.33%, on six hijackers among 72 domains. Given the number of categories and how few hijackers most of them hold, we do not read the ordering below adult as meaningful.

4.4 Hijacking Techniques

Table 5 gives the mechanisms, separating the behaviors a visitor can see from the signals that only the History API exposes. The visible rows carry the same names as the manual taxonomy in Table 1, so the two can be read against each other. Shares add to more than 100% because a page may show more than one.

Mechanism depends on the category. The mix of mechanisms is not the same in every category. Adult sites send the visitor to an unexpected destination in 58.8% of cases against 30.0% pooled and call no History API method on the back press in any of the 68 we confirmed; gambling and gaming sites do the reverse, calling `pushState` or `replaceState` on the press in 51.4% against 15.7%; and news sites hold most of what we cannot explain, with 56.2% carrying no observed mechanism against 30.0%. Appendix B gives the full breakdown.

One mechanism dominates. The same-page trap accounts for 54.1% of hijacking pages: the page pushes an extra history entry, so a back press returns the cursor to a position the visitor has already been in and the URL does not change. Our manual review



(a) Presses needed to reach the referrer, across 1,924 page-level hijacking cases.

(b) CrUX rank distribution for the 364 confirmed hijacking websites.

(c) Categories of the 364 confirmed hijackers.

Figure 3: Characteristics of back-button hijacking websites

Table 5: Mechanisms from the per-press record, split by visibility to the visitor. A page may show several, so columns do not sum to 100%. The unit of *n* differs by row: 2,458 hijacking pages with a readable per-press record, 2,450 the same less eight where history state was unreadable, 1,924 page-level hijacking verdicts, 1,738 hijacking measurements, and 1,625 pages where the press-level API deltas were computable. Only rows sharing a value of *n* are comparable.

	When	Hijacking	Benign	*n*
Visible to the visitor				
Same-page trapping	press	54.1%	—	2,458
Unexpected navigation destination	press	35.5%	—	2,458
Multi-step backward navigation	press	54.7%	—	1,924
Advertisement vignette	press	0.1%	—	1,738
New window or tab opened	press	0.0%	—	1,625
Visible only through the History API				
Extra history entries	load	79.3%	7.9%	2,458
History state set	load	39.2%	20.0%	2,450
`popstate` fired	press	48.0%	—	2,458
`pushState` called	press	20.1%	—	1,625
`replaceState` called	press	19.0%	—	1,625
No mechanism identified		13.5%	—	2,458
counting redirects as none		42.7%	—	2,458

found the same mechanism in 22 of 33 confirmed sites (67%). A human reviewer on 500 domains drawn from the top one million and an automated detector on the top ten thousand agree within four percentage points on which mechanism dominates.

History API use on its own is not a technique. About half of hijacking measurements show `pushState` used somewhere on the page, and about half show a `popstate` handler firing. We list these apart from mechanisms because a single-page application routes with exactly the same calls, so their presence does not identify hijacking on its own. Benign pages do set history state at a comparable rate (20.0% against 39.2%); they plant extra entries far less often (7.9% against 79.3%), which is why we treat the load-time signal as a mechanism and these two as ambient. A detector keyed on History API use would flag a large share of modern websites, which is why we classify on the destination reached instead.

Advertisement interstitials are rare in the back sequence. The advertisement vignette fired in one measurement out of 1,738. We report this plainly because it affects how the behavior should be described. Industry accounts present back-button hijacking as an advertising tactic [9], and our definition treats advertisements as an attribute rather than a criterion (Section 3.2); had we used advertising as the criterion we would have found almost nothing.

One timing detail bears on this. Google AdSense announced in May 2026 that it would stop serving vignette advertisements on the back button from June 15, the same date the spam policy took effect [9]. Our crawl postdates both, so the near absence of vignettes is consistent with that change having already taken hold, and our figure should not be read as the rate before it. We are also cautious about how far the measurement goes: our detector identifies one specific interstitial pattern, so advertising may be present in forms it does not catch.

Some cases have no identifiable mechanism. In 332 of the 2,458 hijacking pages (13.5%) the back press failed to return the visitor and the record names no mechanism, return the visitor and the record names no mechanism, rising to 42.7% if a redirect to an unvisited page is not counted as a mechanism (Table 5); the three figures differ in denominator and in how redirects are treated, not in what was observed. These are not a leftover category: they are the cases our detector cannot settle, and they are why the manual validation in Section 3.3 is necessary rather than confirmatory. We give them their own row rather than spreading them across the named mechanisms.

4.5 Geographic Differences

We first compare the prevalence of back-button hijacking across the eight vantage points. As shown in Table 3, the prevalence ranges from 2.46% to 2.79%, a spread of 0.33 percentage points across locations. We do not claim the rates are equal; the comparison below is at the level of individual websites.

To further investigate geographic variation, we focus on the 6,762 websites successfully measured from all eight vantage points, ensuring that missing measurements are not treated as benign. Among these websites, 247 confirmed websites exhibited back-button hijacking at one or more vantage points. As shown in Figure 4, 93 websites (37.7%) exhibited hijacking at all eight vantage points, while 58 (23.5%) exhibited hijacking at only one vantage point. The

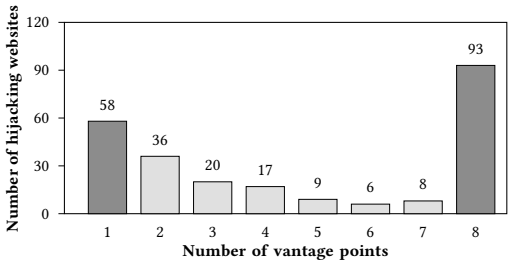


Figure 4: Number of vantage points at which each of the 247 confirmed websites measurable everywhere hijacked.

remaining 96 websites exhibited hijacking at two to seven vantage points.

Among the 58 websites exhibiting hijacking at only one vantage point, 16 were detected in India, approximately twice as many as at any other vantage point. However, India also had the highest proportion of untestable measurements (22.6%) and the lowest internal-page coverage (56.8%).

Similar prevalence at every vantage point therefore conceals two populations, and a crawl from one location would have reported the flat rates and none of the variation behind them.

5 Discussion

In this section, we discuss the implications of our findings for browser security, website operators, and policy enforcement. We also address the ethical considerations and limitations of our measurement study.

5.1 Implications for Browser Security

Chromium skips history entries created without user interaction [2], but websites can still interfere with backward navigation using standard APIs: same-page trapping accounts for 54.1% of hijacking pages, leaving users on the same page after a Back press. The dominant signal precedes the press, and a browser can read it for itself: 79.3% of hijacking pages have already planted an extra history entry at load, against 7.9% of benign pages.

Protections should therefore consider navigation outcomes rather than History API calls or user gestures alone. A browser could detect repeated Back presses that fail to return the user and offer a way past the unwanted entries, though it would have to tell deceptive navigation apart from the routing a single-page application performs with the same calls.

5.2 Implications for Website Operators and Policy Enforcement

Our findings show that back-button hijacking is not limited to website homepages and can vary across geographic locations. Our detection approach can help website operators identify and assess navigation interference across different pages and browsing environments, regardless of whether the behavior is intentionally implemented or introduced by third-party components. Given Google's policy against back-button hijacking [19], website operators should ensure that their websites preserve users' ability to navigate backward as expected.

Classifying on navigation outcomes also makes automated detection possible at this scale, which is what enforcement would need. Our evaluation found both false positives and false negatives, however, so results require validation before they drive enforcement decisions: measurement artifacts and differences between browsing environments both produce records that resemble hijacking.

5.3 Ethical Considerations

Our study investigates back-button hijacking on publicly accessible websites through automated browser measurements. We limited our interactions to website navigation and did not intentionally access authenticated or private content. To reduce the impact on website operations, we used a limited number of concurrent browser instances and isolated browser profiles for each measurement.

Our measurements did not involve human participants. We recognize that identifying websites exhibiting back-button hijacking may have reputational consequences, particularly given Google's policy against this practice. We therefore report aggregate findings rather than publicly identifying individual websites.

5.4 Limitations

First, our detection approach relies on backward navigation outcomes, which may be affected by factors unrelated to back-button hijacking. Despite our checks for measurement artifacts, some cases may still be incorrectly classified. In particular, we could not identify the underlying mechanism in 332 of 2,458 hijacking pages (13.5%). Furthermore, our review of detected websites did not cover negative cases, leaving the possibility of undetected hijacking. Among the 417 detected websites, 364 were confirmed and 53 classified as benign. For 32 of the 364 no screenshot was recorded, so they were never inspected and we count them with the confirmed set, giving a confirmed-positive fraction of 87.3%; counting them against the detector instead gives 79.6%. Second, our measurements were limited to desktop Chrome, with a single visit from a search engine. Different entry paths, returning visits, mobile browsers, or longer interactions may trigger different behaviors. Accepting cookie consent banners may also affect the observed navigation behavior. Moreover, valid internal-page measurements were obtained for only 56.8–61.2% of websites across vantage points, while differences in VPN exit locations and geolocation may influence our geographic comparisons. Finally, our large-scale measurements were conducted after Google's back-button hijacking policy took effect on June 15, 2026. While our findings provide a post-enforcement baseline, future longitudinal measurements could help investigate how back-button hijacking evolves over time.

6 Conclusion

We measured back-button hijacking across the CrUX top 10,000 from eight vantage points and confirmed it on 364 of 9,090 testable websites (4.00%). The behavior is spread evenly across popularity ranks. Computing rates within each category leaves one category distinguishable: adult sites hijack at 5.72% against an overall 4.16% within the classified subset, and adult stays above the overall rate when domains are collapsed to operators. Internal pages account for

29.1% of cases and would be missed by a homepage-only crawl. The dominant mechanism is visible before the user presses anything: 79.3% of hijacking pages carry extra history entries at load, against 7.9% of benign pages. Advertisement interstitials appeared in 1 case of 1,738, so the behavior is a navigation problem rather than an advertising one.

Our measurements were taken after Google's enforcement began on 15 June 2026 and give a post-enforcement baseline. They cannot show whether the policy changed anything, since that needs measurements on both sides of the date using one detector. The wider difficulty is telling deceptive navigation apart from ordinary history management, and our results give browser vendors and search engines a concrete signal to work from.

References

[1] botdetection2026 2026. Detecting Bot Detection: Prevalence, Techniques, and Implications for Web Measurement Research. arXiv:2606.14525. https://arxiv.org/abs/2606.14525 Accessed: 2026-10-17.

[2] Chromium blink-dev. 2019. PSA: History Manipulation Intervention. Google Groups — blink-dev mailing list. https://groups.google.com/a/chromium.org/g/blink-dev/c/T8d4_BRb2xQ Accessed: 2026-10-17.

[3] Nurullah Demir, Matteo Große-Kampmann, Tobias Urban, Christian Wressnegger, Thorsten Holz, and Norbert Pohlmann. 2022. Reproducibility and Replicability of Web Measurement Studies. In *Proceedings of the ACM Web Conference 2022 (WWW '22)*. ACM, New York, NY, USA, 533–544. doi:10.1145/3485447.3512214

[4] Steven Englehardt and Arvind Narayanan. 2016. Online Tracking: A 1-Million-Site Measurement and Analysis. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS '16)*. ACM, New York, NY, USA, 1388–1401. doi:10.1145/2976749.2978313

[5] Devin Ersoy, Brandon Lee, Ananth Shreekumar, Arjun Arunasalam, Muhammad Ibrahim, Antonio Bianchi, and Z. Berkay Celik. 2026. Investigating the Impact of Dark Patterns on LLM-Based Web Agents. In *Proceedings of the 46th IEEE Symposium on Security and Privacy (S&P '26)*. IEEE. https://arxiv.org/abs/2510.18113

[6] Google Chrome Team. 2017. *Chrome User Experience Report (CrUX)*. Technical Report. Google. https://developer.chrome.com/docs/crux Accessed: 2026-10-17.

[7] Colin M. Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L. Toombs. 2018. The Dark (Patterns) Side of UX Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3173574.3174108

[8] Colin M. Gray, Cristiana Teixeira Santos, Nataliia Bielova, and Thomas Mildner. 2024. Towards a Shared Language of Dark Patterns: Harmonizing Taxonomies and Building an Ontology. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM. doi:10.1145/3613904.3642802

[9] Sara Guaglione. 2026. WTF is back button hijacking? Digiday. https://digiday.com/media/wtf-is-back-button-hijacking/.

[10] Johanna Gunawan, Aditya Pradeep, David Choffnes, Woodrow Hartzog, and Christo Wilson. 2021. A Comparative Study of Dark Patterns Across Web and Mobile Modalities. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), 1–29. doi:10.1145/3479521

[11] Umar Iqbal, Charlie Wolfe, Charles Nguyen, Steven Englehardt, and Zubair Shafiq. 2022. Khaleesi: Breaker of Advertising and Tracking Request Chains. In *31st USENIX Security Symposium*.

[12] Soheil Khodayari, Thomas Glauber, and Giancarlo Pellegrino. 2025. Do (Not) Follow the White Rabbit: Challenging the Myth of Harmless Open Redirection. In *Proc. Network and Distributed System Security Symposium (NDSS)*.

[13] Martin Koop, Erik Tews, and Stefan Katzenbeisser. 2020. In-Depth Evaluation of Redirect Tracking and Link Usage. *Proceedings on Privacy Enhancing Technologies* 2020, 4 (2020), 394–413. doi:10.2478/popets-2020-0079

[14] Benjamin Krumnow, Hugo Jonker, and Stefan Karsch. 2022. How Gullible Are Web Measurement Tools? A Case Study Analysing and Strengthening OpenWPM's Reliability. In *Proceedings of the 18th International Conference on Emerging Networking EXperiments and Technologies (CoNEXT '22)*. ACM, New York, NY, USA, 171–186. doi:10.1145/3555050.3569131

[15] Victor Le Pochat, Tom Van Goethem, and Wouter Joosen. 2021. A Practical Guide to Studying the (In)security of Popular Websites. In *Passive and Active Measurement Conference (PAM 2021)*. Springer, 402–418. doi:10.1007/978-3-030-72582-2_24

[16] Arunesh Mathur, Gunes Acar, Michael J. Friedman, Eli Lucherini, Jonathan Mayer, Marshini Chetty, and Arvind Narayanan. 2019. Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (2019), 1–32. doi:10.1145/3359183

[17] Arunesh Mathur, Mihir Kshirsagar, and Jonathan Mayer. 2021. What Makes a Dark Pattern... Dark? Design Attributes, Normative Considerations, and Measurement Methods. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, 1–18. doi:10.1145/3411764.3445610

[18] Célestin Matte, Nataliia Bielova, and Cristiana Santos. 2020. Do Cookie Banners Respect my Choice? Measuring Legal Compliance of Banners from IAB Europe's Transparency and Consent Framework. In *IEEE Symposium on Security and Privacy (S&P)*. 791–809.

[19] Chris Nelson. 2026. Back-Button Hijacking — New Spam Policy. Google Search Central Blog. https://developers.google.com/search/blog/2026/04/back-button-hijacking Accessed: 2026-10-17.

[20] Midas Nouwens, Ilaria Liccardi, Michael Veale, David Karger, and Lalana Kagal. 2020. Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating Their Influence. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, 1–13. doi:10.1145/3313831.3376321

[21] OWASP. [n. d.]. Testing for Reverse Tabnabbing. Web Security Testing Guide. https://owasp.org/www-project-web-security-testing-guide/.

[22] Ali Rasaii, Ha Dao, Anja Feldmann, Mohammadmahdi Javid, Oliver Gasser, and Devashish Gosain. 2025. Intractable Cookie Crumbs: Unveiling the Nexus of Stateful Banner Interaction and Tracking Cookies. *Proceedings on Privacy Enhancing Technologies 2025* 4 (2025), 429–445.

[23] Nelson Rweso, Patrick Meehan, Thomas Reschenhofer, and Schahram Dustdar. 2023. Through the Lens of Google CrUX: Dissecting Web Browsing Experience Across Devices and Countries. *arXiv preprint arXiv:2308.06409* (2023). https://arxiv.org/abs/2308.06409

[24] Iskander Sanchez-Rola, Igor Santos, and Davide Balzarotti. 2019. The Web Is Not Chrome: A Study of Website Testing across Different Browsers. In *Proceedings of the Internet Measurement Conference (IMC '19)*. ACM, 36–49. doi:10.1145/3355369.3355587

[25] Michael Smith, Craig Disselkoen, Shravan Narayan, Fraser Brown, and Deian Stefan. 2018. Browser History re:visited. In *12th USENIX Workshop on Offensive Technologies (WOOT)*.

[26] Sergej Stafeev and Giancarlo Pellegrino. 2024. SoK: State of the Krawlers — Evaluating the Effectiveness of Crawlers for Security and Privacy Measurements. In *Proceedings of the 33rd USENIX Security Symposium (USENIX Security '24)*. USENIX Association. https://www.usenix.org/system/files/sec24summer-prepub-198-stafeev.pdf

[27] 2017. Analyzing the Ecosystem of Malicious URL Redirection Through Longitudinal Observation from Honeypots. *Computers & Security* (2017). https://www.sciencedirect.com/science/article/pii/S016740481730007X Accessed: 2026-10-17.

[28] Kurt Thomas, Elie Bursztein, Chris Grier, Grant Ho, Nav Jagpal, Alexandros Kapravelos, Damon McCoy, Antonio Nappa, Vern Paxson, Paul Pearce, Niels Provos, and Moheeb Abu Rajab. 2015. Ad Injection at Scale: Assessing Deceptive Advertisement Modifications. In *Proceedings of the 36th IEEE Symposium on Security and Privacy (S&P '15)*. IEEE, 151–167.

[29] WHATWG. 2024. HTML Living Standard — Navigation and Session History. Living Standard. https://html.spec.whatwg.org/multipage/nav-history-apis.html Accessed: 2026-10-17.

[30] Xinyu Xing, Wei Meng, Byoungyoung Lee, Udi Weinsberg, Anmol Sheth, Roberto Perdisci, and Wenke Lee. 2015. Understanding Malvertising Through Ad-Injecting Browser Extensions. In *Proceedings of the 24th International Conference on World Wide Web (WWW '15)*. ACM. doi:10.1145/2736277.2741630

[31] Apostolis Zarras, Alexandros Kapravelos, Gianluca Stringhini, Thorsten Holz, Christopher Kruegel, and Giovanni Vigna. 2014. The Dark Alleys of Madison Avenue: Understanding Malicious Advertisements. In *Proceedings of the 2014 Internet Measurement Conference (IMC '14)*. ACM, New York, NY, USA, 373–380. doi:10.1145/2663716.2663719

A Vantage-Point Validation

Validating the vantage points. A comparison between countries depends on each machine being where we claim it is, and this is easy to get wrong without leaving any trace in the data. On most of our hosts the exit address differs from the server named in the VPN configuration, so reading the configuration establishes which server was dialed, not where traffic emerges. We therefore read the live exit at launch and again at every chunk boundary, and record every reading alongside the measurements it covers.

Checking the country alone is insufficient, for two reasons we encountered directly. A tunnel can fail while leaving the machine in the correct country: our German vantage point was at one stage reaching the web through our own institutional network, which is

also in Germany, so every country check passed while the vantage point was no longer a VPN vantage at all. The same domain list gave 6.11% through the institutional network against 3.94% through the tunnel, so the distinction changes the result. We therefore also check the exit organization and reject our own network. Separately, two vantage points in the same country cannot be distinguished by country code, so we record the serving datacenter as well.

Tunnels can also stop carrying traffic while the VPN service continues to report itself healthy. From inside a run this appears as an unreadable exit, so the crawler retries, restarts the tunnel, rechecks the country, and halts the run if the exit remains unreadable. Records whose provenance is unknown are worth less than no records. Across the eight completed vantage points no run triggered a tunnel repair and none halted, so every chunk was crawled from a verified exit in the intended country.

For three target countries the provider serves addresses registered to that country from infrastructure located elsewhere, and geolocation providers disagree about them: two place these addresses in the target country, while the large content-delivery network whose country header much of our corpus receives places them in the routing hub. We report these vantage points with the disagreement stated rather than selecting one answer, since the question of which country such an address belongs to has no answer that all consumers of the address agree on.

B Mechanism by Website Category

Table 6 cross-tabulates the 364 confirmed hijacking websites against the mechanism observed on them; the 34 the classifier declined appear as *Unclassified*. Each website contributes one row, taken at the vantage point whose record held the strongest mechanism evidence, so a site that redirects in one country and traps in another is counted once under whichever of the two ranked higher. The shares are therefore not comparable with Table 5, which counts pages.

We separate two kinds of unexplained case. *No mechanism observed* means the back press failed to return the visitor and the record names no cause, which is a result. *Signal not readable* means the instrumentation could not establish what happened, which is a missing value. Reporting them as one row would overstate the first: pooled, they are 33.0% and 9.4%.

Across the five categories with at least 20 websites the mix differs markedly. The largest differences from the pooled shares are:

- Adult, unexpected destination, 40 of 68 = 58.8% against 30.0% pooled.
- Adult, API call on the back press, 0 of 68 against 15.7%.
- Adult, no mechanism observed, 7 of 68 = 10.3% against 30.0%.
- Gambling and gaming, API call on the back press, 18 of 35 = 51.4% against 15.7%.
- News, unexpected destination, 3 of 48 = 6.2% against 30.0%.
- News, no mechanism observed, 27 of 48 = 56.2% against 30.0%.

The concentration of unexplained cases in news is the one that bears on Table 5. Spread evenly they would look like detector noise; concentrated in news they are consistent with a redirect chain at load, since news sites commonly redirect between locale, scheme or an accelerated-page host before the visitor interacts. The reading is weaker than it appears, because news is also where our instrumentation failed most often, at 20.8% against 9.4% pooled, so part of the concentration is our own blind spot rather than the sites' behavior. We treat these as the differences large enough to describe and interpret no cell beyond them. The twelve categories below 20 websites are listed for completeness and are too thin to support any comparison.

Table 6: Confirmed hijacking websites by category and mechanism, one row per website. Counts, not percentages.

	Visible to the visitor		API	Unexplained		
Category	**Dest.**	**Planted**	**On press**	**None**	**Unread**	***n***
Adult	40	17	0	7	4	68
News	3	6	2	27	10	48
Gambling and gaming	5	1	18	11	0	35
Shopping	6	6	9	13	1	35
Entertainment	11	3	5	7	5	31
Government	2	1	4	5	4	16
Other	2	3	3	5	1	14
Technology	1	5	1	6	1	14
Sports	2	0	0	8	3	13
Finance	1	3	4	2	1	11
Downloads	3	0	1	5	1	10
Social	1	3	1	2	0	7
Travel	4	1	1	1	0	7
Business	1	2	0	3	0	6
Education	2	0	2	2	0	6
Weather	0	2	1	3	0	6
Health	0	0	1	2	0	3
Unclassified	9	4	8	2	11	34
Total	**93**	**57**	**61**	**111**	**42**	**364**